Service Placement Lab

Optimization Methods / Assessment benchmark / Java 17 / 17 October 2026 / version assessment-srs-3.0

A constrained placement simulation with a shared evaluator, Smart Restart Search (SRS), an Exact reference and seeded experimental evidence. The original weighted-latency project is preserved as a separate mode; its scores are never combined with this benchmark.

1. Model and fixed dataset

Decision: one host index A-D for each of 12 indivisible services. There are 4^12 = 16,777,216 complete vectors. Every service is placed exactly once. CPU, RAM, disk, GPU, maximum service count and individual network SLAs are hard rules. Assessment mode has no minimum-two-services rule.

Server	vCPU	RAM GB	Disk GB	GPU	Max jobs	GI/s
A	16	32	160	0	4	12
B	24	48	250	0	5	20
C	32	64	400	2	5	34
D	20	40	240	1	4	24

Server	Idle/peak W	USD / busy s	RTT ms	Bandwidth Mb/s
A	70 / 150	.003	18	500
B	95 / 210	.005	30	800
C	140 / 290	.010	60	1200
D	115 / 235	.007	42	650

The twelve service demands, work GI, GPU needs, payload MB and SLA limits are preloaded from the assessment and editable/resettable in Configuration. Both original CSVs remain unchanged; assessment CSVs live in data/assessment/.

Minimize fixed dimensionless J

J = 0.35(M/30 s) + 0.20(C/0.50 USD) + 0.15(E/5 Wh) + 0.30(L/70 ms)

B[j] = sum Work/Speed in seconds; M = max B. Active power P = Idle + (Peak-Idle)*CPU utilization if busy, else zero. E = sum P*B/3600 Wh; C = sum Price*B USD. Network latency = RTT + 1000*8*payloadMB/bandwidthMbPerSecond ms; L is its mean and Lmax its maximum. Compute/wait time is excluded from the network SLA. The reference scales are normalizers, not hard limits.

2. Method and reproducibility

Smart Restart Search

SRS extends the existing GRASP-style method. It builds a constrained start, improves it through moves/swaps and restarts at a local optimum. GPU eligibility and network SLAs are checked while constructing, and the complete shared evaluator remains the authority for accepting a candidate.

Step	Action / why
1	Rank services by host eligibility and resource pressure, with seeded random variation. Restricted services are placed early.
2	Choose randomly among good eligible hosts with enough remaining CPU/RAM/disk/GPU/service slots and passing SLA. A blocked partial start is abandoned and counted separately.
3	Inspect moves of one service and swaps of two; accept the best strict feasible improvement. Rejected trials never replace the incumbent.
4	When no neighbor improves, restart. Compare global best at each complete batch. Stop at patience, cap or user cancellation.

Defaults: seed 2026, cap 20,000 complete evaluations, batch 10,000, patience 1, greediness 0.35. Greediness, batch and patience are configurable algorithm parameters. Every repeated and infeasible complete evaluation counts. Construction failures are separate. One iteration means one complete evaluation, not a GA generation.

Comparators and proof

Exact enumerates every vector once and can prove the minimum for the same finite model. Random draws 100,000 uniform vectors. Target stops at feasible J <= 0.8 or 500,000 attempts. SRS returns best found unless a completed matching Exact reference establishes equality to the optimum.

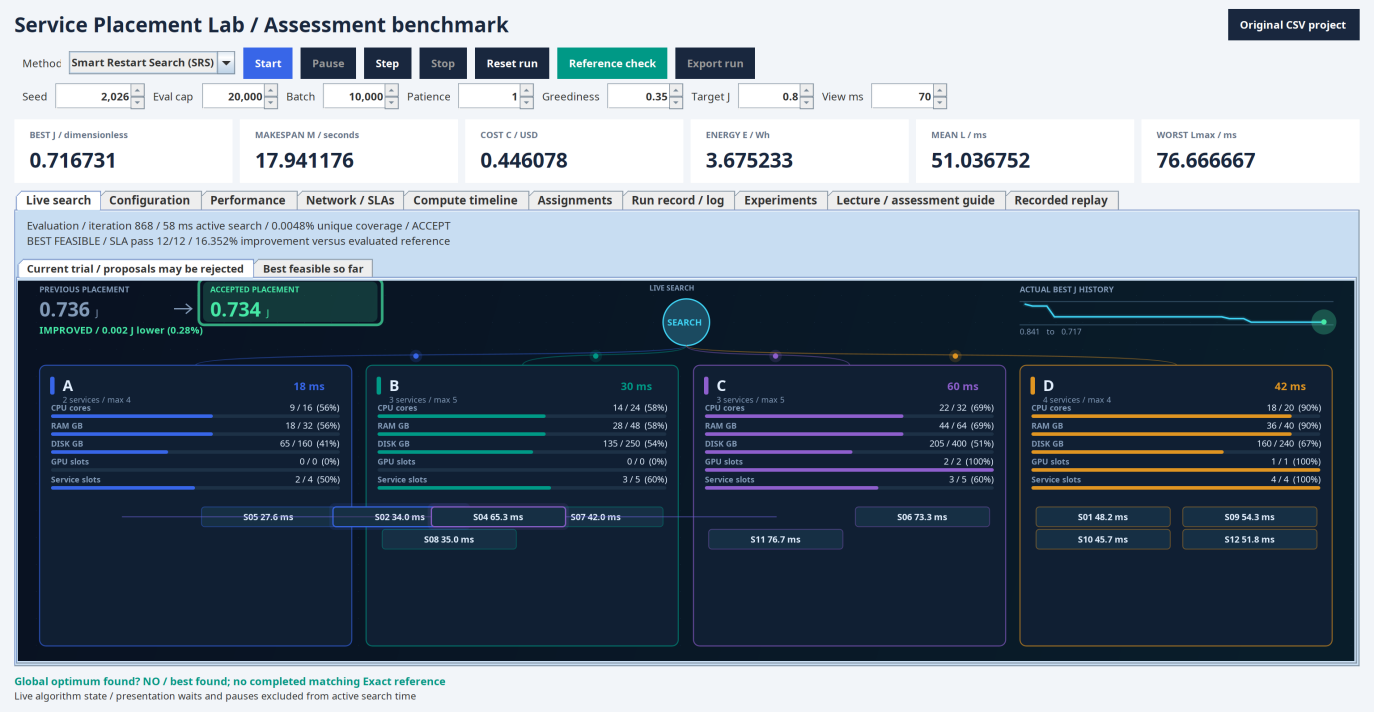
Live controls and runtime

Pause blocks before the next real evaluation; Step permits one and then waits again; Resume releases the search. Current and best canvases come from the same actual event. A separate recorded replay samples real decisions. Presentation pacing/pauses are tracked separately and excluded from active search time; this time is different from simulated makespan and network ms.

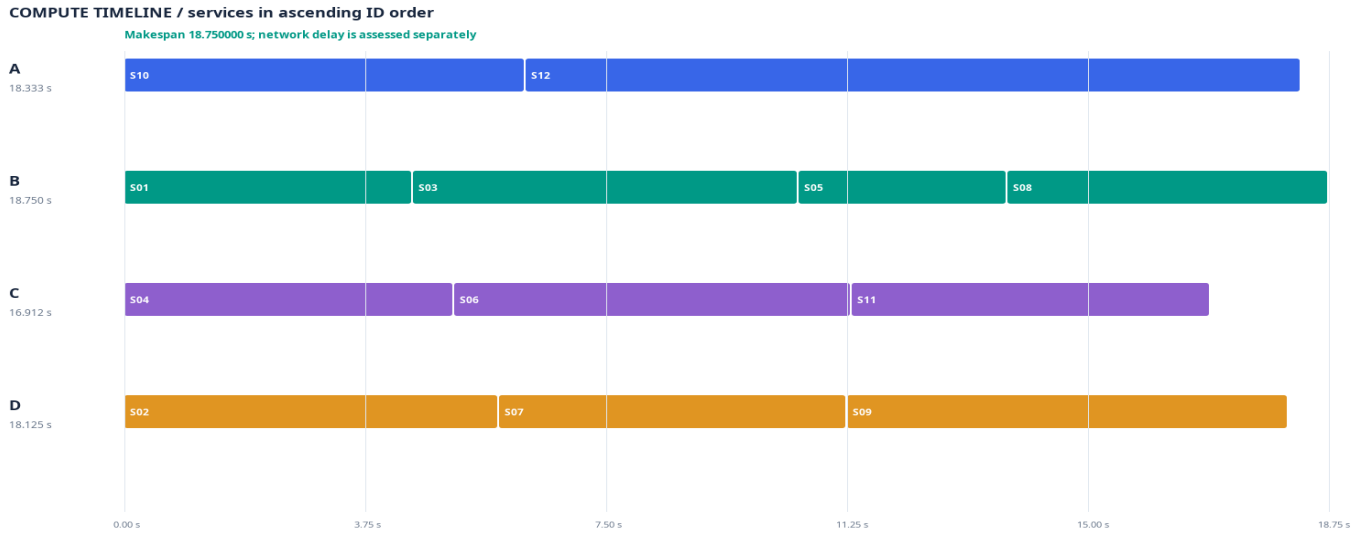
Every run records inputs, instance, version, method, settings, stop reason, seed or Exact=N/A, feasibility, J/M/C/E/L/Lmax/SLA count, evaluation/distinct counts, Java/OS and runtime. Same seed, data, version and settings reproduce results/counts; runtime varies. No external Java libraries or cloud accounts are used.

3. Interface and execution evidence

The current trial and best feasible tabs show resource reservations, GPU/service slots and per-service network latency. Pulses report a numerical score change; failures also have text reasons. Animation moves service chips, while assessment resource numbers remain the exact evaluated reservations.



The timeline uses ascending service ID and Work/Speed durations. Busy-time totals and the makespan are shared with the evaluator; network latency is displayed separately. Six performance panels plot the best-J incumbent and its associated J/L/M/E/C/Lmax measures. Only J must be monotone.



Screenshots were rendered from the actual Java components at 200% scaling. Configuration, network/SLA tables, timeline, statistics and all primary controls were inspected at 1536x795 and 1280x740 logical sizes. PNG/CSV exports preserve the measurements independently of playback.

4. Verification and reference checks

The required T1-T12 scenarios are automated in AssessmentTest and run with the existing model/search/GUI/replay checks. The current suite passed 676 assertions. The log and independent-enumeration evidence are included under docs/evidence/.

Test	Verified behavior
T1	Default returns a feasible best; every recorded accepted incumbent re-evaluates feasible.
T2	S06 on A rejected: GPU requirement 1, capacity 0.
T3	S04 added to A with S01/S03/S07 demands 21 vCPU >16; rejected.
T4	Disable C GPUs: total demand 3 > available 1; necessary-condition infeasibility.
T5	Demand exactly equal to a capacity is accepted.
T6	Identical seed/settings/data reproduce assignment and evaluation/distinct counts.
T7	Zero speed or negative capacity is rejected before searching.
T8	Ordered timeline totals equal B[j]; maximum equals M.
T9	S03 on C =70 ms >60 ms SLA; rejected regardless of J.
T10	Zero bandwidth or negative RTT rejected before searching.
T11	Lower A bandwidth with fixed feasible mapping raises its service latencies, L and J.
T12	All reference quantities match PDF rounding; 12/12 network SLAs pass.

Reference mapping: A={S01,S03,S07}, B={S12}, C={S02,S05,S06,S10,S11}, D={S04,S08,S09}. This is an evaluated numerical self-check, not a search result or optimum claim.

J	M seconds	C USD	E Wh	L ms	Lmax ms
0.856845	28.3333	0.449583	3.927416	53.347863	76.666667

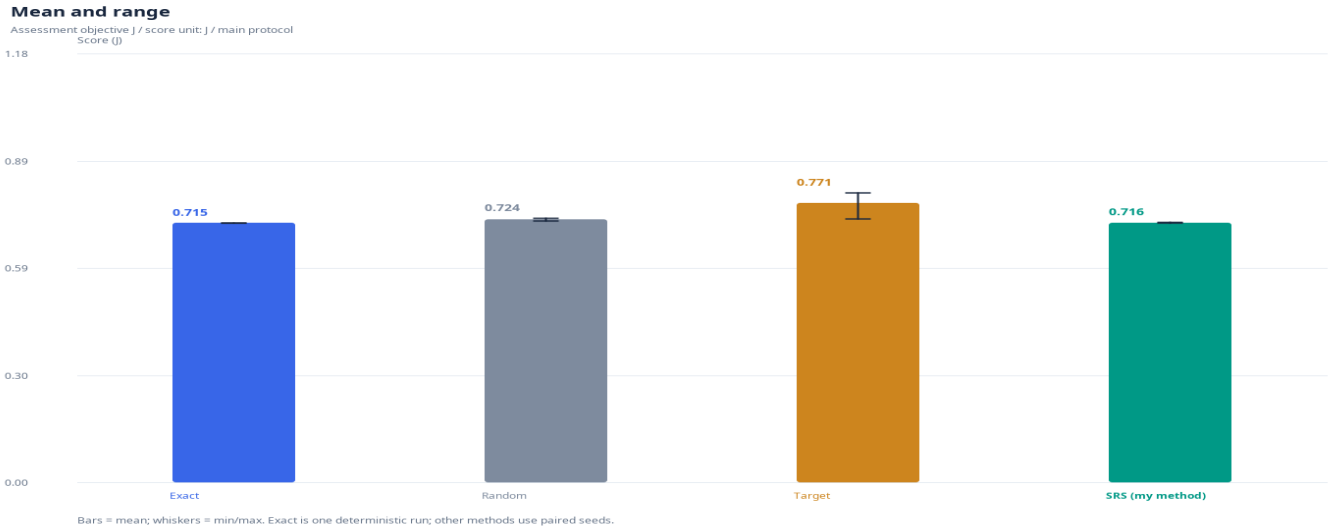
An independent recursive checker confirmed 92,643 feasible assessment placements and the Exact minimum 0.7151543595803707. Verification checks implementation of the equations; validation of physical response times would require real measurements and stakeholder review.

5. Repeated experiments and budget

Twenty paired seeds 2026-2045; Exact once with no seed. Main protocol: Random=100,000, Target<=0.8 or 500,000, SRS=20,000/batch10,000/patience1. All returned main results are feasible; target success is distinct from matching Exact. J is dimensionless.

Method	Best J	Mean J	Worst J	Exact hits	Mean evals	Mean ms
Exact Search	0.715154	0.715154	0.715154	100%	16,777,216	2397.00
Random Search	0.721222	0.724076	0.728393	0%	100,000	96.65
Target Search	0.727454	0.770779	0.797635	0%	469	0.25
SRS	0.715154	0.716057	0.716358	25%	20,000	12.35

Exact minimum: M=18.5 s, C=0.444950980392 USD, E=3.684492442810 Wh, L=49.188034188034 ms, Lmax=76.666666666667 ms; all SLAs pass. SRS main mean is about 0.1262% above Exact, using 0.119209% of full-space size in evaluation calls. Its unique coverage is smaller because trials can repeat.



At equal 100,000 evaluations, SRS mean J=0.715274687343 and 18/20 hit Exact; Random mean J=0.724076291660. At matched SRS stop counts (20,000), Random mean J=0.727881183774. A single lucky run is not used as the evidence.

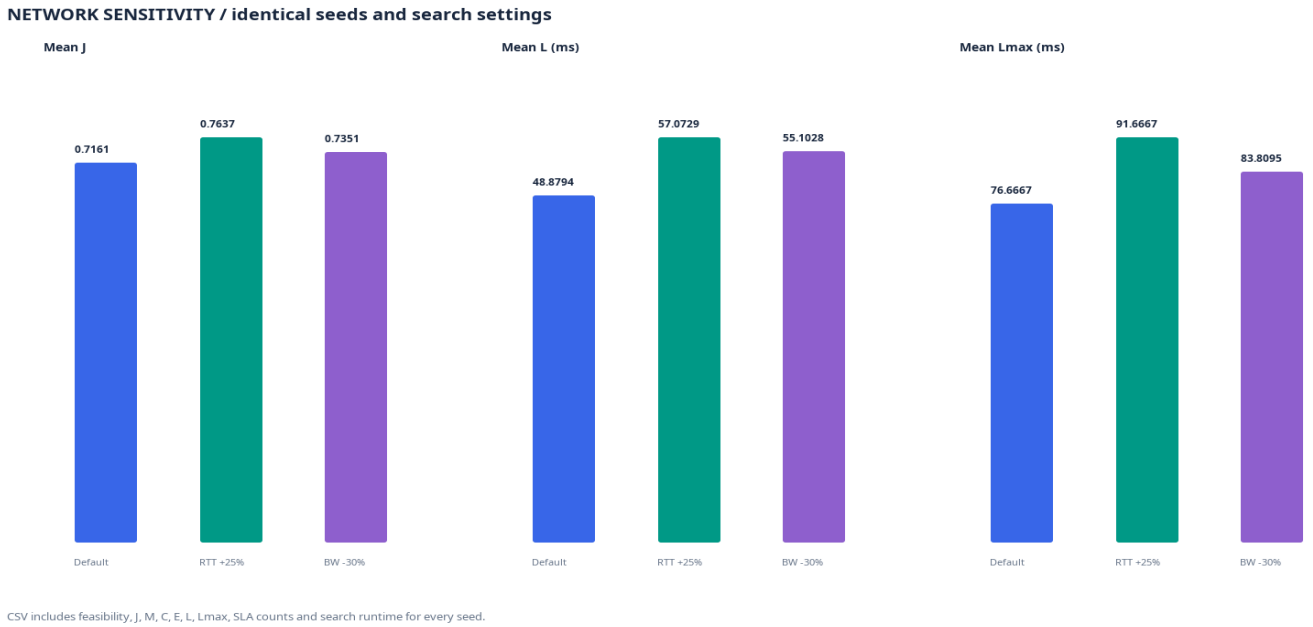
Budget (evals)	Mean J	Best J	Sample SD J	Feasible
20	0.759449	0.739492	0.013892	20/20
50	0.757675	0.739492	0.012366	20/20
100	0.741299	0.726488	0.008340	20/20
200	0.734309	0.716358	0.008006	20/20

Budget sensitivity keeps other settings fixed and disables patience. Means improve as the cap rises from 20 to 200 complete evaluations. All full statistics, seed dots, fair-budget comparisons and traces are exported as CSV/PNG. Runtime is machine/JVM-specific; this is descriptive evidence, not a universal ranking.

6. Network sensitivity, limits and credit

Use the same twenty seeds and SRS settings. Change all RTTs by +25%, or all bandwidths by -30%; keep other data fixed. The following are feasible heuristic results, not certified sensitivity optima.

Scenario	Mean J	Mean L ms	Mean Lmax ms	Feasible
default	0.716057	48.879380	76.666667	20/20
RTT +25%	0.763720	57.072863	91.666667	20/20
Bandwidth -30%	0.735109	55.102778	83.809524	20/20



Communication deterioration raises L/Lmax and mean J, sometimes changing which placements pass SLAs. Three required GPU slots consume all C/D GPU capacity, restricting those services. Throughput, busy-time price and power affect load/energy/cost; optimizing the combined J can trade one component against another.

Limits and tested scope

This is the specified static simulation: no transfer contention, packet loss, inter-service traffic, replicas or real cloud operation. A failed search alone is inconclusive; total-resource necessary conditions or complete enumeration can prove infeasibility. SRS remains approximate even when a particular result matches Exact.

Linux Java 17 compilation, native launch, automated controls and 200% component rendering were tested. Windows/macOS, a physical projector and an IntelliJ GUI session were not directly tested. No screen recording is claimed; the supplied five-minute script supports a live demonstration.

Sources, reuse and assistance

Sources: Assessment_Latency.pdf and the supplied Optimization Methods Week 0/1 lectures. The student supplied the original Java idea/data; this extends the existing project. OpenAI Codex assisted implementation, tests and documentation. A temporary Java formatter and an independent Python checker were development tools only. No external Java runtime libraries are required. The student must review/explain the code and credit assistance under institutional rules.

Evidence: results/assessment-study/ contains 121 main/fair records plus 80 budget and 60 network records (261 total), with full input snapshots. The separate original weighted project and its illustrative trade-off sweep remain available; their scores are never compared numerically with J.